

Natural Language Processing, HW3

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Q1. Method Description (5%)

- Clearly describe your overall retrieval pipeline and the main idea behind your method.

Ans:

We built a multi-stage hybrid retrieval pipeline that combines lexical matching (BM25), semantic dense retrieval (BGE-M3), cross-encoder reranking, and LLM-based listwise reranking (Qwen3-8B).

Stage	Component	Input	Output
1	BM25 Retrieval	Question + all candidates	Ranked candidate list
2	Dense Retrieval (BGE-M3)	Question + all candidates	Ranked candidate list
3	RRF Fusion	BM25 + Dense ranked lists	Fused ranked list (all candidates)
4	Cross-Encoder Reranker (BGE-Reranker)	All fused candidates	Reranked top-10
5	LLM Listwise Reranker (Qwen3-8B)	Top-10 + modality hint	Final top-5 ranking
6	Adaptive Modality Quota	Top-5 + evidence_modality_type	Final submitted top-5

The main idea is to combine the complementary strengths of lexical, semantic, and generative ranking, while using the dataset's own metadata (evidence_modality_type) as a hard constraint to bypass the ranking problem entirely for a large subset of samples.

- Explain the retriever, evidence preprocessing, ranking procedure, and any additional techniques you used.

Ans:

1. Retriever

- BM25

Lexical matching using lowercased, punctuation-stripped

tokenization. Effective for exact term matches such as company names, financial metric names, and fiscal years.

- Dense (BGE-M3):

Multilingual dense embedding model. Query is prefixed with "query: " following BGE-M3 instruction format. Cosine similarity computed via dot product on L2-normalized vectors.

2. Evidence Preprocessing

- Text quotes: used as-is from the text field.
- Image quotes: the `img_description` field is used as the text representation.

3. Ranking Procedure

- Reciprocal Rank Fusion (RRF)

Scores from BM25 and Dense rankings are fused using RRF with $k=60$: $\text{score}(d) = \sum 1/(k + \text{rank}(d))$. This is applied over the complete candidate set without truncation, ensuring no candidate is discarded before the cross-encoder stage.

- Cross-Encoder Reranking

All fused candidates are passed to a BGE-Reranker cross-encoder, which jointly encodes the question and each candidate to produce a relevance score. Unlike the bi-encoder in Stage 2, the cross-encoder can model fine-grained interactions between question tokens and evidence tokens. The top-10 candidates by cross-encoder score are forwarded to the LLM stage. Passing all candidates (rather than a truncated top-k) to the cross-encoder is critical: analysis of the training set showed that 34% of missed gold answers in mixed-modality samples were ranked 11th–15th by BM25, meaning a `RETRIEVAL_K=10` cutoff would permanently discard them.

- LLM Listwise Reranking (Qwen3-8B)

The top-10 cross-encoder candidates are passed to Qwen3-8B (loaded in 4-bit quantization via BitsAndBytes) as a listwise reranking task. The model receives the question, a numbered list of candidate texts (truncated to 400 characters each), and an expected modality hint derived from `evidence_modality_type`. It is asked to output a JSON array of candidate indices ordered by relevance.

4. Additional Techniques

The additional technique is the use of the `evidence_modality_type` field as an explicit signal to steer the final candidate selection, which we refer to as Adaptive Modality Quota.

Q2. Comparison of Retrieval Methods (5%)

- Based on your experimental results, compare the performance of BM25, Dense Retriever, direct LLM selection, and your own method.

Ans:

Method	Recall@5(Kaggle test set)
BM25	0.65485
Dense Retriever	0.684
LLM Selection	0.636
Our Method	0.8128

- Analyze the strengths and weaknesses of all four methods, and explain what you think causes the performance differences.

Ans:

Method	Strengths	Weaknesses
BM25	- No GPU required. - Reliable for exact keyword matches. - Stable and predictable output.	- Cannot capture semantic meaning. - Systematically underranks image candidates.
Dense Retriever	- Captures semantic relationships beyond exact keyword matching. - Encodes both text and image descriptions into the same vector	Image candidates still suffer from a semantic gap between the question and <code>img_description</code> text.

	space for unified ranking.	2. Requires GPU for reasonable speed.
LLM Selection	1. Strong language understanding in theory. 2. No need for separate embedding or reranking models.	1. Unstable output format causes parsing failures. 2. Slow inference speed.
Our Method	1. Cross-encoder captures token interactions; 2. modality-aware selection	1. Higher latency and compute cost 2. Requires LLM inference per sample

The core reason for the performance ordering is the progressive addition of complementary signals:

- BM25 provides strong lexical precision that LLM selection lacks.
- Dense retrieval adds semantic coverage that BM25 misses.
- Cross-encoder reranking adds cross-candidate interaction that bi-encoders cannot model.
- Adaptive modality quota eliminates the single largest source of systematic error: allocating prediction slots to the wrong modality.

Q3. Multimodal Embedding vs. Text-Description Retrieval for Image Evidence(10%)

- There are two common ways to retrieve image evidence :
 - a. Use a multimodal embedding model to encode the image directly into a vector, then compare it with the question embedding.
 - b. Convert each image into a text description, then retrieve it together with other text evidence using a text-only retriever. You may either use the `img_description` field that is already provided in the dataset, or generate your own description with a VLM / image-captioning model.
- Briefly state which option you chose and why.

Ans:

We chose option (b). We used the `img_description` field provided in the dataset rather than generating our own descriptions with a VLM. This decision was made for two reasons. First, it allows all candidates both text quotes and image quotes to be encoded in the same vector space, enabling direct and fair comparison through cosine similarity. Second, it avoids the additional computational cost and complexity of running a separate VLM for description generation.

- Compare your experimental results for these two approaches. Discuss which one performs better on this task and explain what you think causes the difference.

Ans:

Method	Kaggle Score
Text-Description (bge-m3)	0.684
Multimodal Embedding (bge-visualized-m3)	0.371

For this task, text-description retrieval is clearly the better approach. The modality gap in multimodal embedding models causes image candidates to be systematically underranked, which is especially harmful given the high proportion of image-based ground-truth answers. By converting images to text descriptions and using a unified text encoder, we avoid the modality gap entirely and achieve substantially better retrieval performance.

Q4. Modality Preference Analysis (10%)

- Examine whether your retriever shows a preference for a particular modality in its retrieval results.
 - For example, does it tend to retrieve more text evidence, or does it favor image evidence?
 - Based on your actual experimental results, describe this preference and analyze what you think causes it.
 - Even if your retriever shows no clear preference, you are still expected to explain what aspects of your method cause this balanced behavior.

Ans:

Our retriever shows a clear preference for image evidence. Across all 1,798 test samples, 71.7% of predicted quotes are image quotes, compared to 28.3% text quotes.

1. Primary cause: Adaptive Modality Quota

The dominant driver of the image preference is the Adaptive Modality Quota. For pure-image samples (41.8% of the test set), all 5

prediction slots are forcibly assigned to image candidates, regardless of the LLM's ranking output. This alone accounts for most of the 71.7% image rate:

Expected image predictions from adaptive quota alone: 6,475 (72.1%) — nearly identical to the actual 6,437 (71.7%). This confirms that the image preference is almost entirely explained by the hard constraint in Stage 6, not by any intrinsic bias in the BM25, dense, or LLM components.

2. Secondary cause: Dataset composition

The dataset itself is image-heavy. Among all train gold quotes, 60.2% are image quotes, reflecting the fact that the source documents are financial reports and technical papers where key quantitative evidence appears in tables and charts. Our pipeline correctly tracks this underlying distribution and amplifies it slightly due to the pure-image quota rule.

3. Minor cause: LLM modality hint

For mixed-modality samples, the LLM receives an expected evidence modality hint (e.g., "table, text") in its prompt. This nudges the LLM to rank image candidates slightly higher than it would without the hint, contributing marginally to the image preference in the mixed-sample subset. However, since the effect is constrained by the minimum-2-image rule rather than a hard all-image rule, this contribution is small compared to the pure-image quota.